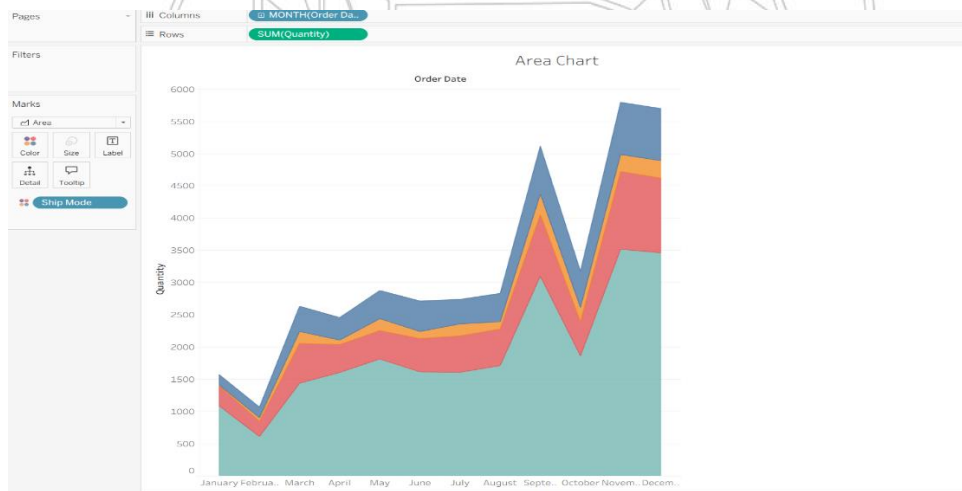


Data Analytics with Tableau Assignment

Sheet: Area Chart

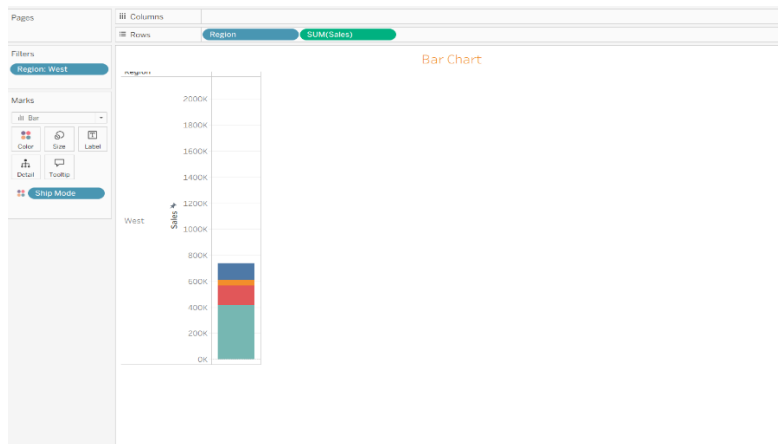
- Start by placing Order Date on the Columns shelf.
- Change the date level from Year to Month.
- Add Quantity to the Rows shelf.
- Use Ship Mode on the Marks card and apply it to Color.
- Change the chart type to Area Chart from the Marks card options.



Sheet 02: Bar Chart

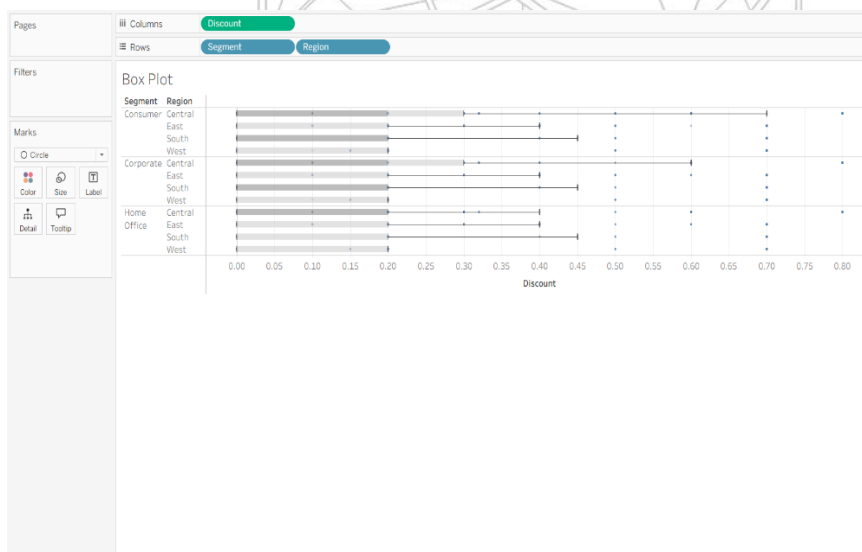
- Place Sales on the Rows shelf to display total sales.
- Add Order Date to the view.
- Convert the default chart into a Bar Chart using the Marks card.
- Add Ship Mode to Color for better visualization.
- Place Region before Sales on the Rows shelf.
- Apply a filter on Region and keep only the West region selected





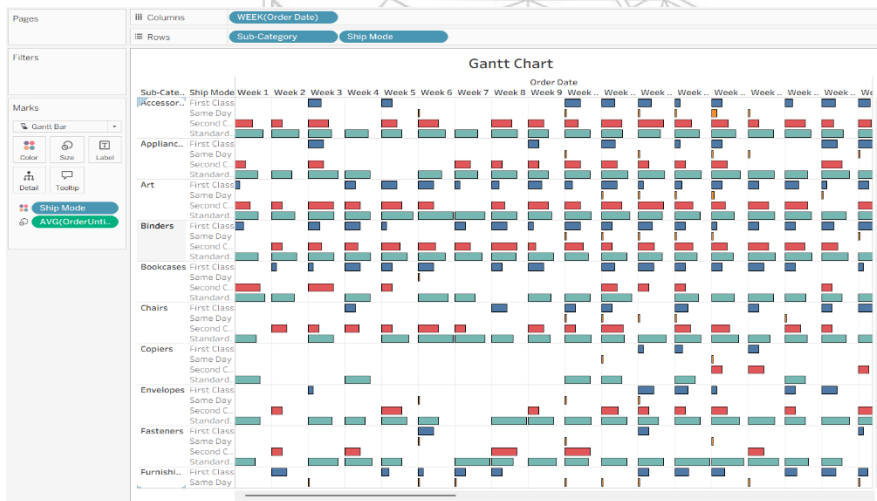
Sheet 03: Box Plot

- Drag Segment to the Columns shelf.
- Place Discount on the Rows shelf and change its calculation to Average.
- Add Region next to Segment.
- Open Show Me and choose the Box-and-Whisker Plot option.
- Turn off Aggregate Measures from the Analysis menu.
- Use the Swap button to switch rows and columns.



Sheet 04: Gantt Chart

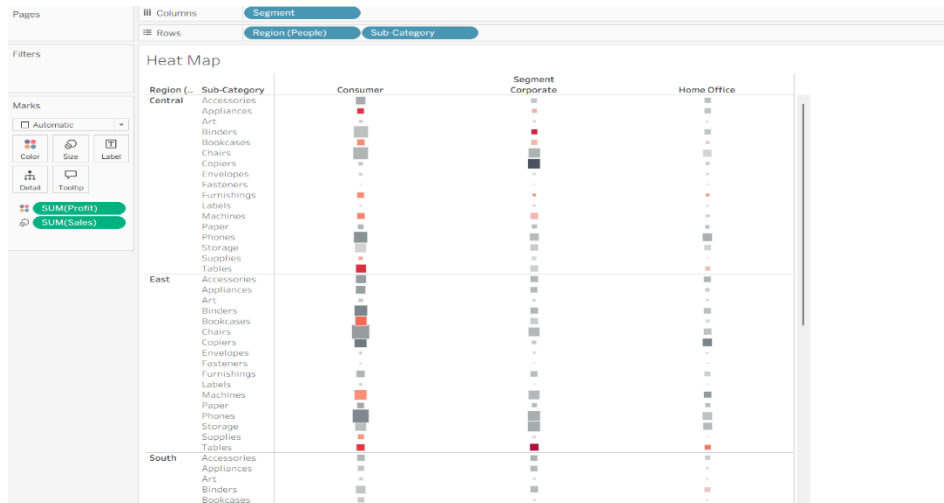
- Add Order Date to Columns and change the date format to Week Number.
- Place Sub-Category and Ship Mode on the Rows shelf.
- Select Gantt Bar as the chart type.
- Create a calculated field named OrderUntilShip using the DATEDIFF() function.
- Drag this field to Size to represent shipping duration.
- Change its aggregation from Sum to Average.
- Use **Ship Mode** as the color field.



Sheet 05: Heat Map

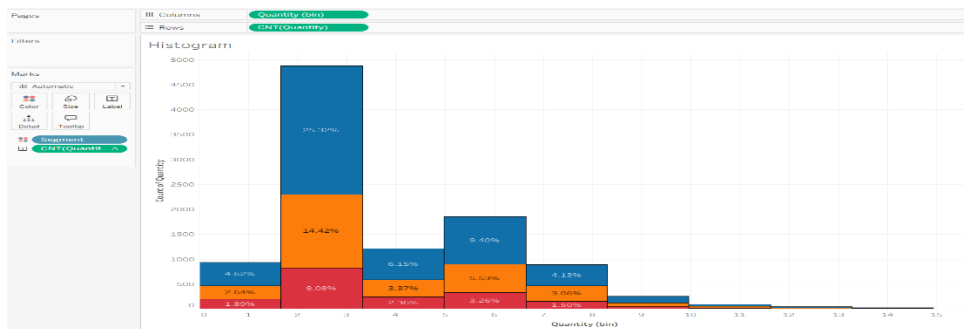
- Place Segment in Columns.
- Add Region and Sub-Category to Rows.
- Use Profit on Color to create the heat map effect.
- Apply a medium gray border from the Color settings.
- Choose an appropriate color palette using Edit Colors.
- Add Sales to Size so larger values appear as larger boxes.
- Adjust the size slider for a clear display.





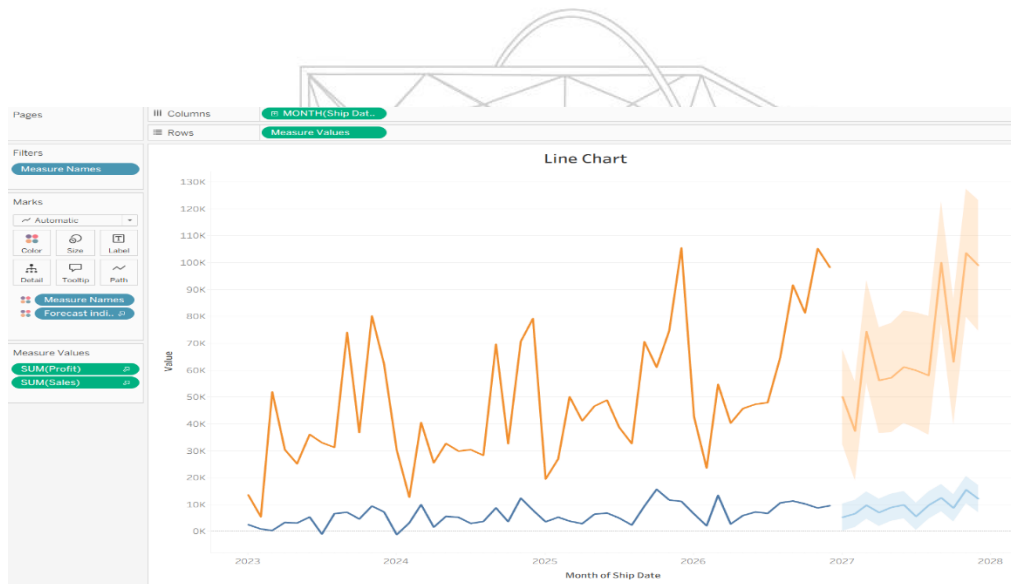
Sheet 06: Histogram

- Drag Quantity to Columns.
- Select Histogram from the Show Me panel.
- Tableau automatically creates bins and counts.
- Add Segment to Color to compare customer segments.
- Copy CNT(Quantity) to Label.
- Apply Percent of Total as a quick table calculation to display percentages.



Sheet 07: Line Chart

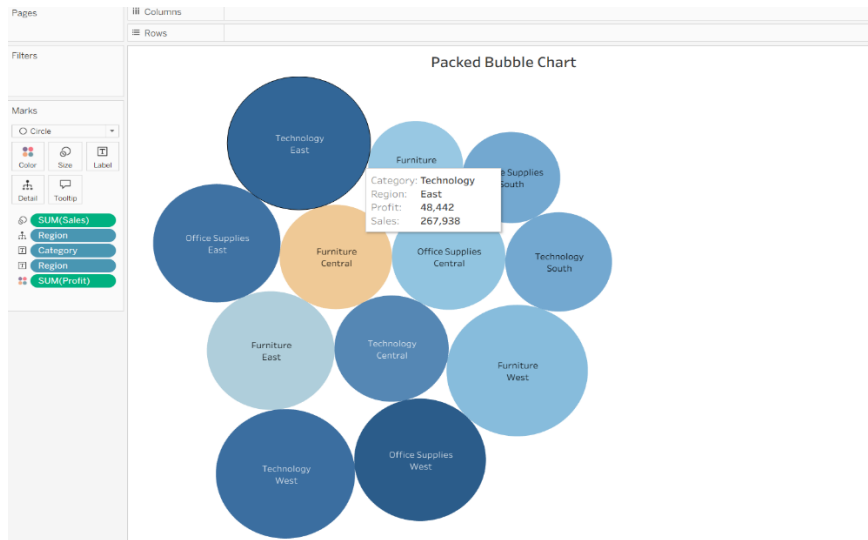
- Place Ship Date on Columns.
- Add Sales and Profit to Rows.
- Combine both measures on a shared axis by dragging Profit onto the Sales axis.
- Change the date level to Year and add a Forecast from the Analytics pane.
- Switch the date level back to Month for detailed analysis.



Sheet 08: Packed Bubbles

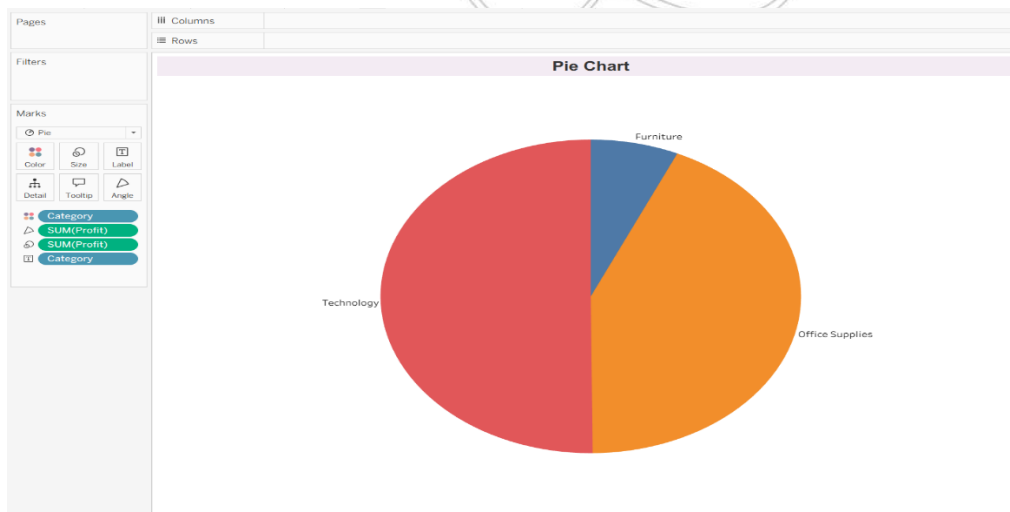
- Add Category and Sales to the worksheet.
- Choose Packed Bubbles from Show Me.
- Place Region on Detail to divide the bubbles further.
- Use Profit on Color.
- Add Region to Label so each bubble displays its region name.





Sheet 09: Pie Chart

- Add Profit and Category to the worksheet.
- Select Pie Chart from Show Me.
- Tableau automatically uses Profit for size and angle.
- Increase the pie size using the Size slider.
- Display category names by adding Category to Label.



Sheet 10: Scatter Plot

- Place Sales on Columns and Profit on Rows.
- Use Category on Color to distinguish product groups.
- Add Region to Detail to create additional data points.
- Insert a Linear Trend Line from the Analytics pane.



Sheet 11: Text Table

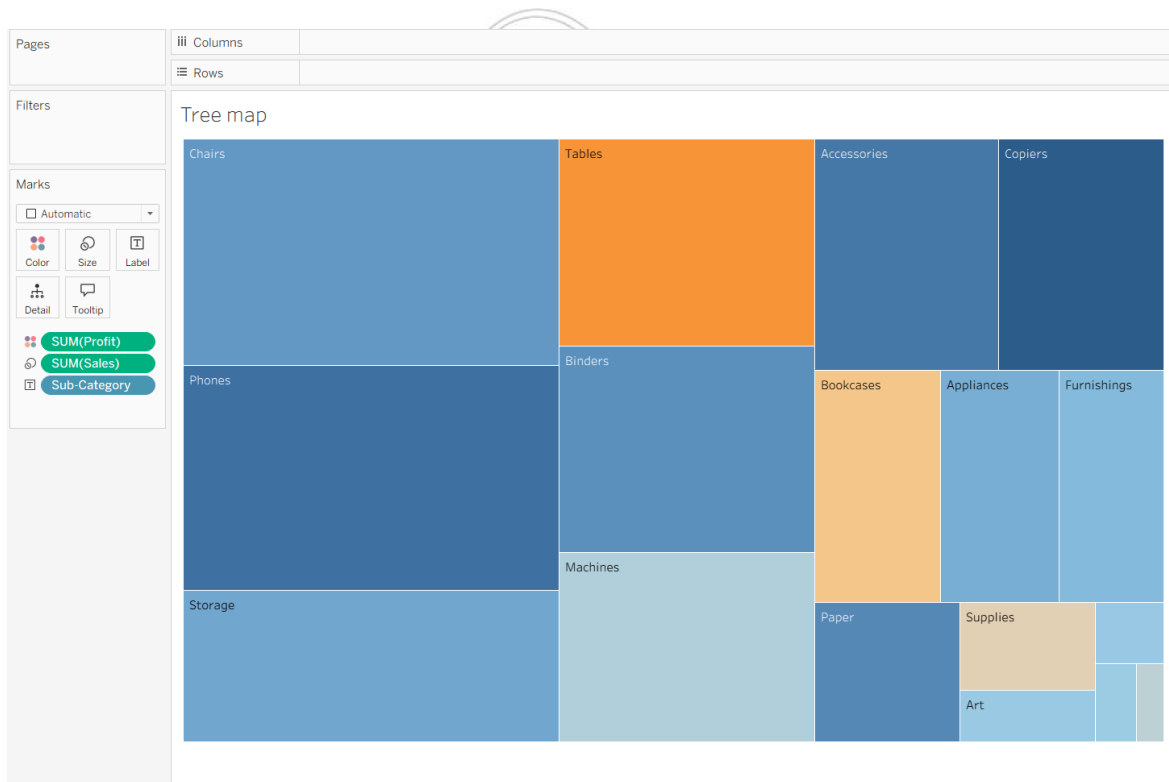
- Add Order Date to Columns and Sub-Category to Rows.
- Display Sales values on Text.
- Insert Region before Sub-Category in Rows.
- Apply a Percent of Total table calculation to Sales.
- Set the calculation to compute using Pane.

Text Table With Percentage

Region	Sub-Category	2021	2022	2023	2024
Central	Accessories	4.77%	7.58%	7.27%	7.53%
	Appliances	3.52%	4.84%	4.05%	6.04%
	Art	0.79%	1.20%	1.02%	1.55%
	Binders	15.98%	5.79%	10.59%	14.51%
	Bookcases	1.77%	0.07%	5.04%	3.01%
	Chairs	19.99%	17.41%	15.74%	16.13%
	Copiers	3.15%	12.45%	11.78%	2.49%
	Envelopes	1.54%	0.80%	0.65%	0.83%
	Fastereners	0.17%	0.09%	0.17%	0.22%
	Furnishings	2.44%	2.40%	3.65%	3.44%
East	Accessories	15.89%	1.80%	1.79%	4.02%
	Appliances	1.01%	0.80%	0.34%	0.40%
	Art	15.89%	1.80%	1.79%	4.02%
	Binders	2.26%	3.43%	3.61%	4.22%
	Bookcases	9.64%	18.82%	15.39%	15.70%
	Chairs	10.68%	0.40%	0.63%	9.28%
	Copiers	0.40%	0.35%	2.87%	2.98%
	Envelopes	7.50%	6.67%	9.37%	7.10%
	Fastereners	4.47%	11.40%	3.43%	6.85%
	Furnishings	4.51%	4.26%	5.22%	5.73%
South	Accessories	1.02%	1.13%	1.11%	2.27%
	Appliances	5.40%	9.18%	10.81%	6.84%
	Art	8.92%	12.51%	3.30%	3.42%
	Binders	16.89%	12.75%	12.70%	14.50%
	Bookcases	5.91%	3.16%	10.81%	11.27%
	Chairs	0.99%	0.55%	0.71%	0.39%
	Copiers	0.14%	0.14%	0.21%	0.15%
	Envelopes	3.32%	3.46%	5.43%	5.07%
	Fastereners	0.38%	0.33%	0.47%	0.42%
	Furnishings	10.81%	9.18%	13.80%	9.46%
West	Accessories	3.25%	2.53%	2.71%	3.62%
	Appliances	16.14%	14.40%	11.46%	16.79%
	Art	12.07%	8.48%	12.11%	9.49%
	Binders	0.36%	0.70%	1.14%	3.25%
	Bookcases	8.96%	5.67%	4.29%	5.48%
	Chairs	5.39%	2.80%	10.02%	6.01%
	Copiers	2.04%	3.40%	3.99%	6.47%
	Envelopes	0.55%	1.91%	1.54%	1.05%
	Fastereners	0.55%	1.91%	1.54%	1.05%
	Furnishings	0.55%	1.91%	1.54%	1.05%

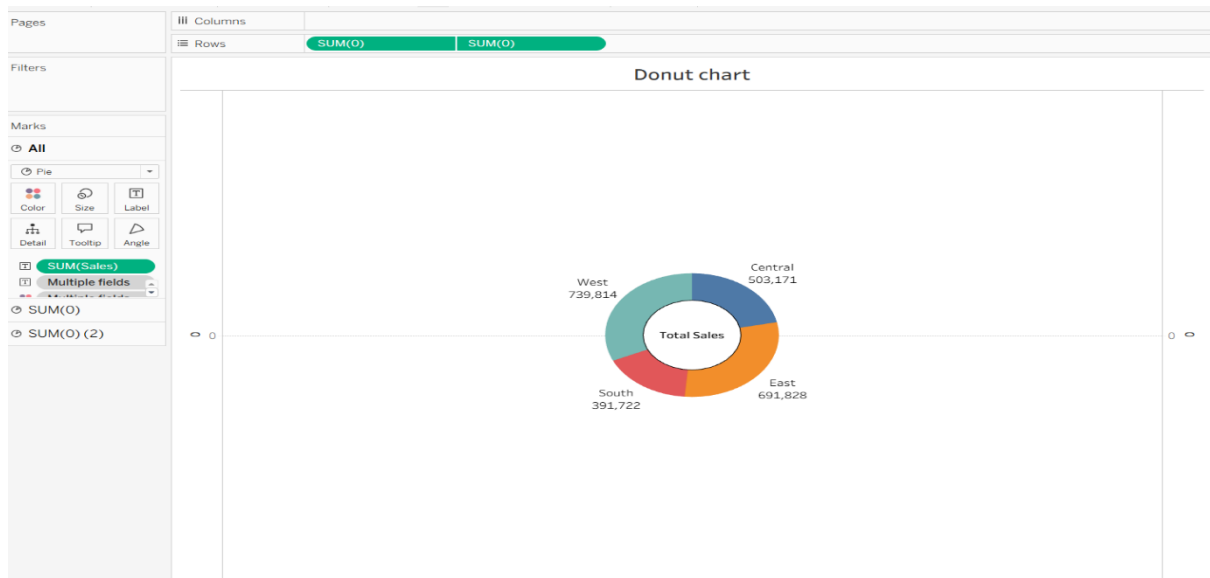
Sheet 12: Tree Map

- Place Sub-Category and Sales in the worksheet.
- Choose Treemap from Show Me.
- Use Profit on Color.
- Rectangle size represents Sales, while color represents Profit.



Sheet 13: Donut Chart

- Create a calculated field called Zero with a value of 0.
- Add Zero twice to the Rows shelf.
- Change the Marks type to Pie.
- Create a Dual Axis to overlap both pie charts.
- Reduce the size of the second pie and color it white.
- Apply Region to Color on the first pie to complete the donut chart.



Sheet 14: Map Chart

- Drag State to the workspace to generate a map.
- Use Sales on Color to represent sales values.
- Select either Map or Filled Map from the Marks card.
- Add State and Sales to Label.
- Apply a Region filter if needed.
- Set the geographic location to United States and review the final sales map.

